QUEENSLAND UNIVERSITY OF TECHNOLOGY

EGH490-2 · Research Project 2 Progress Report

PROGRESS REPORT

Adaptive Control of Solar Power Shufflers for

Maximum Power Point Tracking Under Partial Shading

APPLICATION OF MODERN STATE-SPACE METHODS
TO SERIAL PARTIAL POWER CONVERTERS

STUDENT ENGINEER	SUPERVISOR
Jared Saunders	Assoc Prof Geoff Walker
n11034459	School of Electrical Engineering & Robotics

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1.0	17/04/26	Jared Saunders	Initial Proposal
2.0	29/05/26	Jared Saunders	Semester 1 Progress Report

1. Executive Summary

This report documents the progress achieved during the first semester of the EGH490 capstone project Adaptive Control of Solar Power Shufflers for Maximum Power Point Tracking Under Partial Shading. The project investigates the application of modern state-space control methods to the serial partial power converter (S-PPC) topology, a class of differential power processing converter that mitigates the energy losses associated with partial shading of photovoltaic (PV) strings.

The project objective is to design a universal control system for an S-PPC string operating under partial shading. The intended end-application is a residential-scale PV-battery system; the controller must track the true global maximum power point of a shaded array while delivering stable power to a battery load. Expected outcomes by project completion are a validated mathematical model, a discretised cascaded controller implemented in firmware, a hardware prototype, and an experimental evaluation under representative partial shading scenarios.

Semester 1 has established the analytical foundation and an initial Simulink implementation. The averaged state-space model of the interleaved half-bridge shuffler has been derived from first principles, linearised about the equilibrium operating point, and analysed for stability, controllability and observability. The plant has been discretised at the target switching frequency and an analytical cascaded PI controller has been designed using classical frequency-domain methods. The PV string and shuffler topology have been implemented in MATLAB / Simulink and validated against analytical maximum-power-point predictions across three partial shading scenarios. Closure of the inner phase current loop has been attempted with a fixed current reference; this work has surfaced an instability under mismatched irradiance that is presently the dominant technical blocker (2.10.5).

Significant implementation work remains. The two outer loops (PV-voltage regulation and phase-current balance) are not yet closed in simulation, the QSPICE behavioural-source implementation of the PI controller is not yet operational, and the model does not yet include the MPPT outer loop or the battery load that defines the target end-use. The most significant project change is the deferral of the first PCB iteration from Semester 1 into early Semester 2, captured in Appendix A. The Semester 2 priority is to maintain momentum on these items so the buffer absorbed by the deferral is not extended further.

2. Interim Deliverables and Technical Progress

2.1 Checklist of Work Completed

The control systems work followed the modern state-space design process. Table 1 summarises the activities executed this semester and their current status. The presentation in 2.2 onwards walks through each item, beginning with the converter topology and ending with the inner-loop closure in Simulink and the present technical blocker.

Activity	Status
State-space modelling derivation, equilibrium, linearisation	Complete
Stability, controllability, observability analysis	Complete
Discretisation (ZOH) and analytical cascaded PI design	Complete
PV module characterisation (suitable representation)	Complete

Activity	Status
Simulink PV string model and partial-shading validation	Complete
Inner phase current loop (Loop 1)	In progress
Voltage loop (Loop 2) and phase-balance loop (Loop 3)	Not started
QSPICE PI controller implementation	Not yet working
MPPT integration and battery load model	Deferred to Sem 2
Compare Avg SS Model to Simulink Switching Model to QSPICE Model	Deferred to Sem 2

Table 1. Status of Semester 1 technical activities.

2.2 Power Shuffler Topology

The selected S-PPC topology is the interleaved half-bridge buck-boost converter with coupled inductors, shown in Figure 1. The converter sits between two PV substrings within a series-stacked string, sharing a common midpoint node V_mid. Each half-bridge leg comprises a high-side switch and a low-side switch in series, with each switch-node driving one winding of the coupled inductor. The two inductor windings share the common midpoint, providing the differential power path between the two PV substrings.

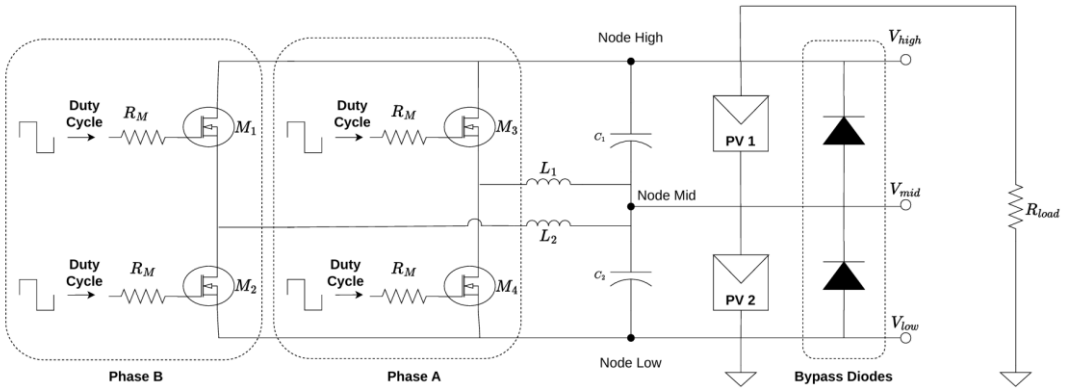


Figure 1. Interleaved half-bridge shuffler topology. Two PV substrings PV1 and PV2 stacked in series across the string output (V_high to V_low). Phase A (M3, M4) and Phase B (M1, M2) each drive one winding of the coupled inductor through their respective switch nodes; C1 and C2 are the substring decoupling capacitors. Bypass diodes are shown for completeness.

Figure 1 fixes the sign and node conventions used throughout the report and in Appendix B: V_C1 is the voltage across the top substring (V_high − V_mid), V_C2 is the voltage across the bottom substring (V_mid − V_low), and the phase currents I_L1 and I_L2 are positive in the direction V_mid → switch node. The load is taken across the full string (V_high to V_low) and represents the downstream battery / MPPT-front-end in the target application.

2.3 Switching State Identification

The interleaved half-bridge has four independently controlled switches, giving $2^4 = 16$ raw switch combinations. Seven of these are forbidden (any combination that simultaneously closes both switches in a single leg shorts that leg through the capacitor). The remaining nine valid states fall into three groups, summarised in Table 2. Only the four dual-switch states are control-relevant the four single-switch states and the all-off state are transient dead-time configurations dynamically equivalent to one of the dual-switch states depending on inductor-current polarity.

Group	Count	Role
Dual-switch (one switch closed per leg)	4	Control-relevant basis of the averaged model
Single-switch (only one switch closed)	4	Dead-time transient, body-diode dependent
All-off	1	Dead-time transient at PWM boundaries

Table 2. Grouping of the 9 valid topological states. The 4 dual-switch states form the averaged model basis (see Appendix B B.3 for the per-state derivations).

2.4 Averaged State-Space Model

The state vector adopted for the analysis is $x = [I_L1, I_L2, V_C1, V_C2]^T$, with the duty cycles of the two phases d_A and d_B as control inputs and the PV substring currents I_pv1, I_pv2 together with the load current I_load as exogenous inputs. An individual state-space representation was derived for each of the four dual-switch states by direct application of Kirchhoff's current and voltage laws. Time-averaging weighted by the dwell fraction of each state d_A·d_B, d_A(1−d_B), (1−d_A)d_B, and (1−d_A)(1−d_B) respectively and assuming continuous conduction mode yields the averaged nonlinear model:

$$L_1\, dI_L1/dt = -d_A\, V_C1 + (1 - d_A)\, V_C2$$

$$L_2\, dI_L2/dt = -d_B\, V_C1 + (1 - d_B)\, V_C2$$

$$C_1\, dV_C1/dt = d_A\, I_L1 + d_B\, I_L2 + I_{pv1} - I_load$$

$$C_2\, dV_C2/dt = -(1 - d_A)\, I_L1 - (1 - d_B)\, I_L2 + I_{pv2}$$

The structural pattern across the four states reduces to a simple rule: capacitor C1 receives current from each phase whose high-side switch is conducting, and C2 supplies current to each phase whose low-side switch is conducting. The input matrix is identical across all four states because the exogenous PV and load currents enter the capacitor nodes independently of switching state. The full per-state derivation, including the assembled bilinear A and B matrices, is in Appendix B B.3–B.4.

2.5 Equilibrium Operating Point

Setting all time derivatives to zero, the volt-second balance on the two inductors yields a structural constraint that the two duty cycles must be equal at steady state:

$$D_A = D_B \equiv D \quad and \quad \bar{V}_C2/\bar{V}_C1 = D/1 - D$$

This is a significant structural result, derived in Appendix B (B.6)–(B.7): the duty cycle sets the voltage ratio between the two PV substrings, which is the degree of freedom that the shuffler exploits to rebalance mismatched modules. The differential duty (d_A − d_B) is a transient control input only and cannot persist at steady state. Solving the capacitor charge-balance equations with the load current included gives the steady-state duty and average inductor current:

$$D = I_load - I_{pv1}/I_load - I_{pv1} + I_{pv2}, \quad \bar{I} = I_{pv2}/2(1 - D)$$

The no-load case exposes a degenerate equilibrium in which all PV currents fall to zero, confirming that a finite load is required for a non-trivial steady state. This is the analytical justification for the load model carried through Appendix B B.5.

2.6 Linearisation About the Operating Point

Small-signal perturbation analysis was performed by substituting x = X̄ + x̃, d = D + d̃, u_pv = Ū_pv + ũ_pv into the averaged equations and retaining first-order terms. The resulting continuous-time linearised model takes the form x̃̇ = Ã·x̃ + B̃_d·d̃ + B̃_e·ũ_pv. The three matrices are shown in Figure 2.

$$\tilde{A} = \begin{bmatrix} 0 & 0 & -\frac{D}{L_1} & \frac{1-D}{L_1} \\ 0 & 0 & -\frac{D}{L_2} & \frac{1-D}{L_2} \\ \frac{D}{C_1} & \frac{D}{C_1} & 0 & 0 \\ -\frac{1-D}{C_2} & -\frac{1-D}{C_2} & 0 & 0 \end{bmatrix}$$

$$\tilde{B}_d = \begin{bmatrix} -\frac{V_\Sigma}{L_1} & 0 \\ 0 & -\frac{V_\Sigma}{L_2} \\ \frac{\bar{I}}{C_1} & \frac{\bar{I}}{C_1} \\ \frac{\bar{I}}{C_2} & \frac{\bar{I}}{C_2} \end{bmatrix} \qquad \tilde{B}_e = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{1}{C_1} & 0 & -\frac{1}{C_1} \\ 0 & \frac{1}{C_2} & 0 \end{bmatrix}$$

Figure 2. Linearised state-space matrices. Ã (top) carries operating-point dependence on D, $L_{1,2}$ and $C_{1,2}$. B̃_d (bottom-left) carries the duty-perturbation gains and depends on V_Σ = V̄_C1 + V̄_C2 and the steady-state per-phase current Ī. B̃_e (bottom-right) is duty-independent.

Figure 2 reveals that the duty perturbations decompose naturally into common-mode and differential-mode components. The common-mode duty ũ_c = (d̃_A + d̃_B)/2 drives the three-dimensional common-mode subsystem comprising I_Σ = I_L1 + I_L2, V_C1, and V_C2 this is where bus voltage regulation and ultimately MPPT live. The differential-mode duty ũ_δ = d̃_A − d̃_B drives the one-dimensional differential current I_Δ = I_L1 − I_L2 through a pure integrator, decoupled from the common-mode dynamics. This decomposition is the analytical basis for the cascaded control architecture in 2.9.

2.7 Stability, Controllability and Observability

2.7.1 Stability

The eigenvalues of Ã were derived analytically by exploiting its rank-deficient structure (rows 1 and 2 are identical when $L_1 = L_2$, as are columns 1 and 2). After modal reduction:

$$\sigma(\tilde{A}) \in \{ 0, 0, +j\omega_0, -j\omega_0 \}, \quad \omega_0 = \sqrt{(2D^2/LC_1 + 2(1-D)^2/LC_2)}$$

Substituting the design values L = 8.2 µH, $C_1 = C_2$ = 100 µF, D = 0.5 gives $\omega_0 \approx 35$ krad/s, corresponding to a natural resonance of approximately 5.5 kHz. The two zero eigenvalues correspond to integrator-like behaviour in the differential current mode and a conserved combination of the common-mode states; the imaginary pair represents an undamped LC oscillation. The open-loop system is therefore marginally stable. This is the expected result for a lossless converter model the controller must provide all damping.

2.7.2 Controllability

The modal decomposition makes the controllability conclusion straightforward: the one-dimensional differential mode is trivially controllable by the differential duty so long as V_Σ ≠ 0, and the three-dimensional common-mode subsystem is generically controllable by the common-mode duty so long as Ī ≠ 0. The plant is therefore fully controllable at any meaningful operating point. Loss of controllability occurs only at three boundary conditions: average inductor current Ī → 0 (the converter enters discontinuous conduction mode); duty cycle saturated to D = 0 or D = 1 (linearisation breaks down handled in implementation by anti-windup); and bus voltage V_Σ → 0 (physically unreachable).

2.7.3 Observability

Observability depends entirely on the chosen sensor set. The observability matrix was constructed for three candidate configurations (Table 3). The critical finding is that capacitor voltage measurements alone yield rank-3 observability the differential current mode is hidden, because the two capacitor voltages depend only on the sum I_L1 + I_L2 through the linearised dynamics. Per-phase current sensing is therefore a non-negotiable requirement for any interleaved implementation. Both phase currents will be measured directly in the hardware design, which removes the need for a state observer in the present project.

Sensor set	Rank of O	Observable?
V_C1, V_C2 only	3 of 4	No differential current is hidden
V_C1, V_C2, I_L1	4 of 4	Yes (minimum viable set)
V_C1, V_C2, I_L1, I_L2	4 of 4	Yes (full state directly measured)

Table 3. Observability of candidate sensor sets. The voltage-only case is rank-deficient (see Appendix B B.9).

2.8 Discretisation

The plant is intended to operate at a switching frequency of 200 kHz, giving T_sw = 5 µs. The control loop is sampled synchronously with the switching frequency to avoid beat artefacts. Discretisation of the plant uses the zero-order hold (ZOH) transformation, which is exact for piecewise-constant duty:

$$A_d = exp(\tilde{A} \cdot T_s), \quad B_d = [object\ Object]^{\wedge}T_s\ exp(\tilde{A} \cdot \tau)\ d\tau \cdot \tilde{B}$$

Because Ã is singular (two zero eigenvalues), the integral form is required and was evaluated via the augmented-matrix exponential. The discrete eigenvalues map to the unit circle: the two zero continuous eigenvalues become unity in discrete time, and the imaginary pair maps to $\cos(\omega_0 T_s) \pm j \sin(\omega_0 T_s) \approx 0.985 \pm j\ 0.174$. The discrete plant inherits the continuous plant's marginal stability, as expected. Controller discretisation will use the Tustin (bilinear) transform when the analogue PI prototypes are converted to digital form.

2.9 Cascaded PI Controller Design

The modal decomposition in 2.6 motivates the three-loop cascaded architecture in Figure 3. The bandwidth allocation follows the standard cascaded-design rule of approximately 10× separation between adjacent loops so each outer loop sees its inner loop as a near-instantaneous tracker.

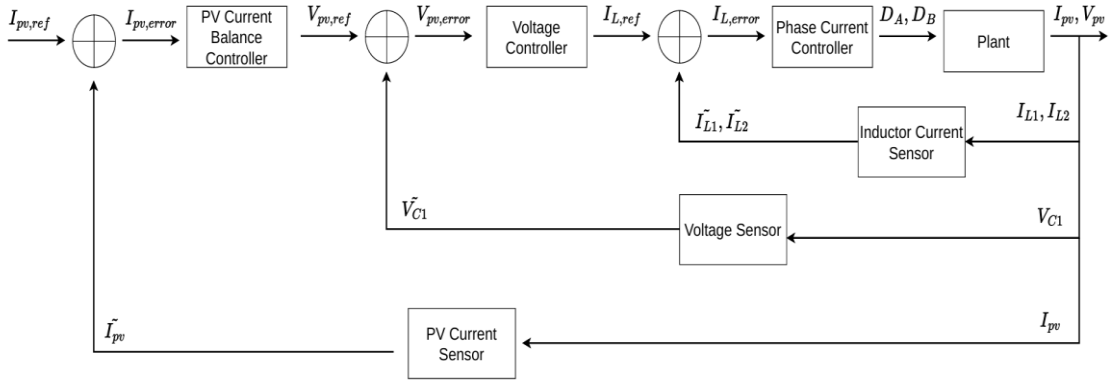


Figure 3. Cascaded three-loop PI architecture. Inner loop (phase current): tracks I_L1 and I_L2 to their references. Middle loop (PV voltage): regulates V_C1 by commanding the inner current reference. Outer loop (PV-current balance): aligns the phase currents on the PV side. An MPPT outer-outer loop will set the V_C1 reference in Semester 2.

Loop	Quantity controlled	Target bandwidth
Inner (per phase)	I_L1, I_L2 (phase currents)	10 kHz
Middle	V_C1 (PV substring voltage)	1 kHz
Outer (balance)	I_pv1 − I_pv2 (PV current differential)	≈ 5 kHz
MPPT (Sem 2)	V_C1 reference setpoint	≈ 10 Hz

Table 4. Cascaded control loop hierarchy with target crossover frequencies.

PI gains for each loop were derived analytically (Appendix B B.11). For the inner current loop the plant is a pure integrator with gain V_Σ / L, giving:

$$K_p,i = \omega_ci \cdot L/V_\Sigma, \quad \tau_i = 10/\omega_ci$$

The same procedure applied to the voltage loop, treating the closed inner loop as ideal, gives K_p,v = ω_cv · C_1 / (2D). All three loops achieve a designed phase margin of approximately 84° from the placement of the PI zero one decade below crossover.

2.10 Simulink Implementation and Validation

2.10.1 PV module representation

Before any string-level work could begin, a suitable representation of an individual PV module was required. The single-diode equivalent model was implemented in Simulink with parameters drawn from the AIKO Neostar 3P60 (P/N: AIKO-A555-MCE60Mw) datasheet, providing a representative module without committing the analytical work to a specific manufacturer. The model was sanity-checked against the MATLAB Solar Cell Power Curve reference example. Figure 4 shows the irradiance-dependent I-V and P-V curves at 25 °C; the open-circuit voltage, short-circuit current, and maximum power point reproduce the datasheet values to within the precision needed for control-design purposes.

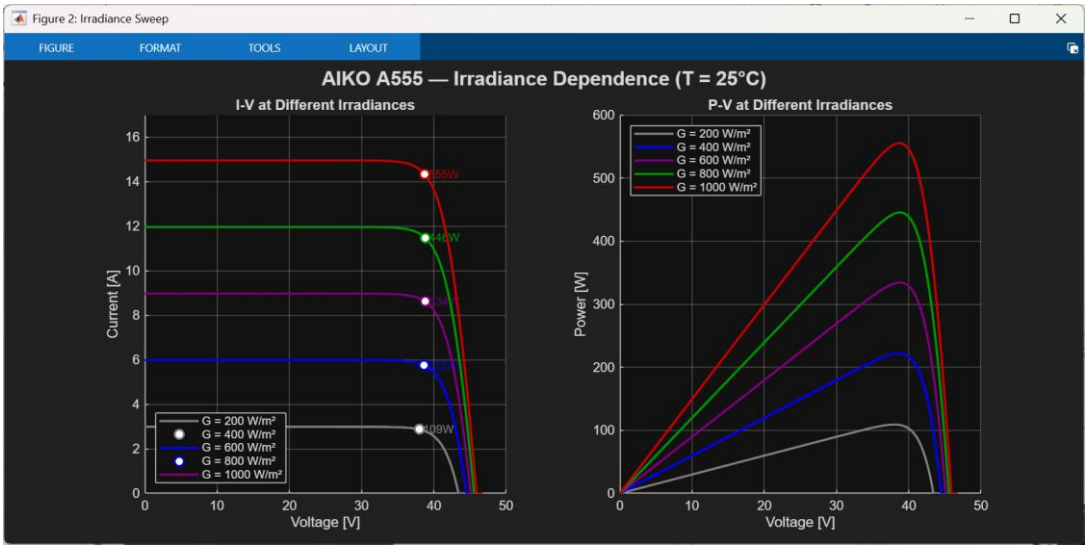


Figure 4. AIKO Neostar 3P60 PV module I-V and P-V curves at five irradiance levels (T = 25 °C). The marker on each I-V curve denotes the maximum power point at that irradiance. This is the per-module representation used for all subsequent string-level work.

The module representation is sufficient for the control-design objective of this semester. The representation does not capture every second-order effect of a real module (notably the temperature dependence of the diode saturation current and the series-resistance variation with irradiance), but the deviations are small relative to the modelling abstractions already accepted in the averaged shuffler model. A more rigorous characterisation against measured cells would be appropriate before the hardware prototype campaign in Semester 2.

2.10.2 PV string model

The PV string was built up incrementally, beginning with a two-module string driven by a ramped controlled-voltage source for I-V curve sweeping. The model was then extended to the full four-module string used for partial shading validation. Figure 5 shows the assembled Simulink model: the four-module string (left), the per-module solar-cell subsystem (centre), and the I-V sweep load used to trace the steady-state P-V curve (right).

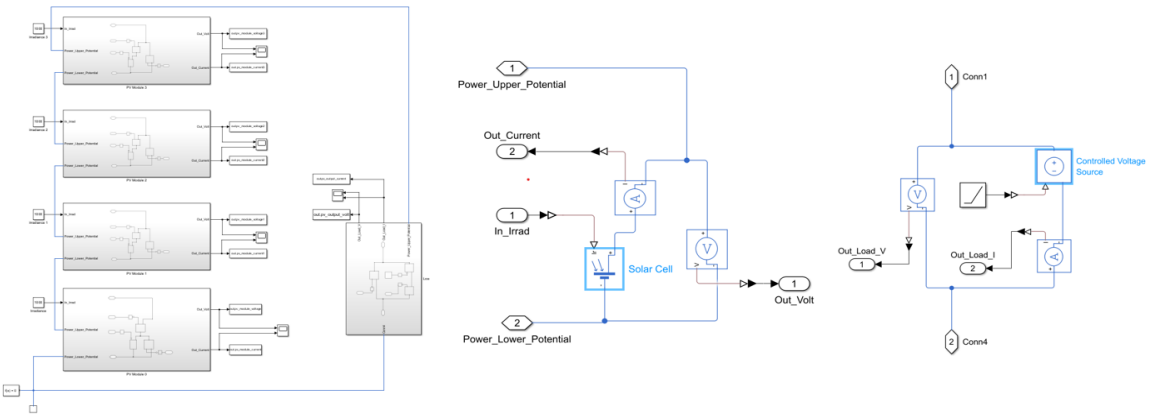


Figure 5. Simulink PV string model. (a) Four-module series-stacked PV string with per-module irradiance and temperature inputs. (b) Internal structure of the per-module solar-cell subsystem (single-diode equivalent). (c) Controlled-voltage-source load used to perform the I-V sweeps that produce the P-V curves in Figure 6.

2.10.3 Partial shading validation

Three partial shading scenarios were used to validate the assembled string model against analytical predictions of the global maximum power point. The irradiance distributions cover the canonical test cases of uniform mismatch, graduated shading, and asymmetric single-module shading. Figure 6 shows the resulting P-V curves; the maximum power point identified by the Simulink sweep agrees with the analytical P_MPP value in all three scenarios (Table 5). Multiple local maxima are visible in the graduated and single-shaded cases this is the local-maximum trapping problem that will motivate the global MPPT design in Semester 2.

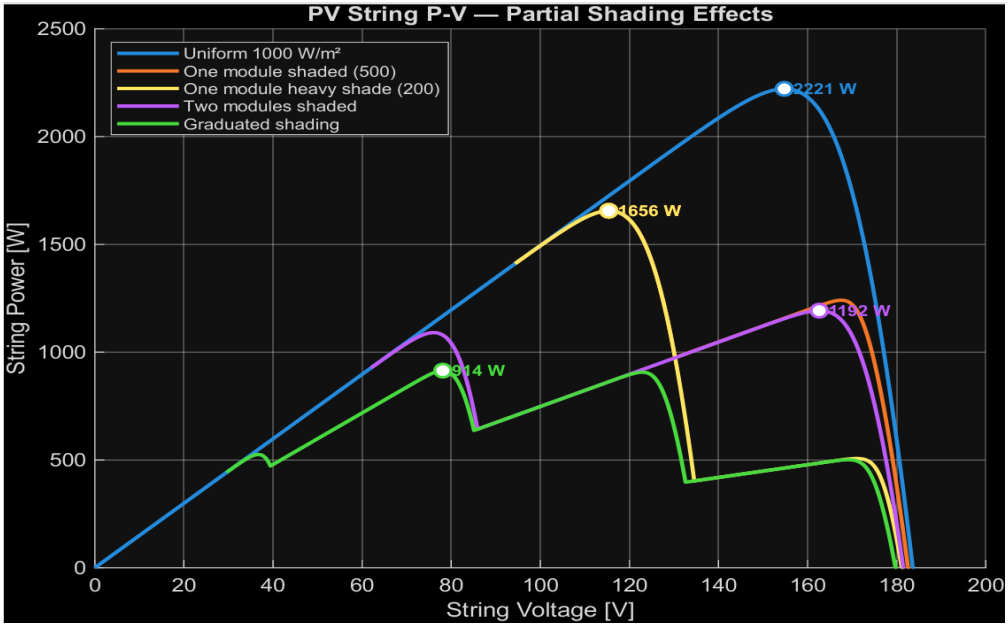


Figure 6. Simulink-swept P-V curves for the partial-shading validation scenarios. Markers indicate the global maximum power point identified for each scenario; for the graduated and single-shaded cases (yellow and green) local maxima are clearly visible at lower string voltages.

Scenario	Irradiance distribution (W/m²)	Analytical P_MPP
Two modules shaded	1000, 1000, 500, 500	1192.5 W
Graduated shading	1000, 800, 500, 200	914.3 W
Single module shaded	1000, 1000, 1000, 500	1655.9 W

Table 5. Partial shading validation scenarios and analytical P_MPP predictions.

2.10.4 Shuffler integration and inner-loop attempt

The interleaved half-bridge shuffler was integrated into the four-module PV string and operated at a fixed 50% duty to verify switching behaviour and the topological assumptions underpinning the averaged model. Three implementation issues required attention: the gating logic for complementary high-side / low-side operation with dead time, the PV module bypass diodes (which interact with the shuffler current path during heavy shading), and the coupled-inductor model (initially approximated by two independent inductors). Figure 7 shows the resulting Simulink shuffler model with the inner phase-current PI loops in place; each phase has its own current sensor, a fixed 3 A reference (the static test value), a PI controller (PID block), and a gate-driver subsystem.

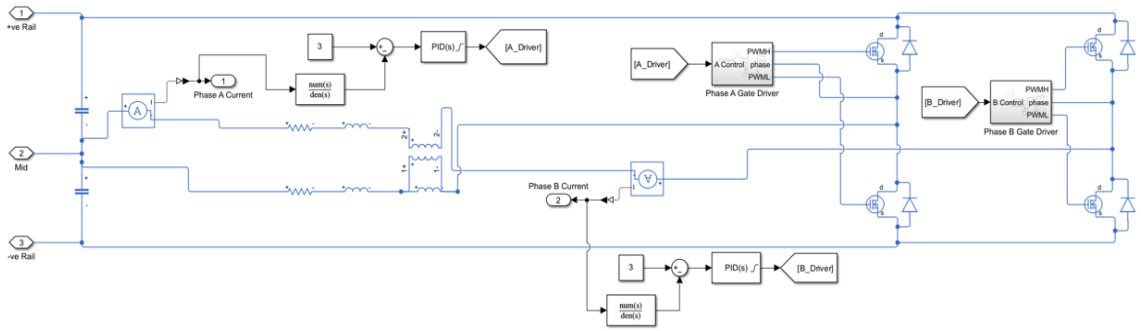
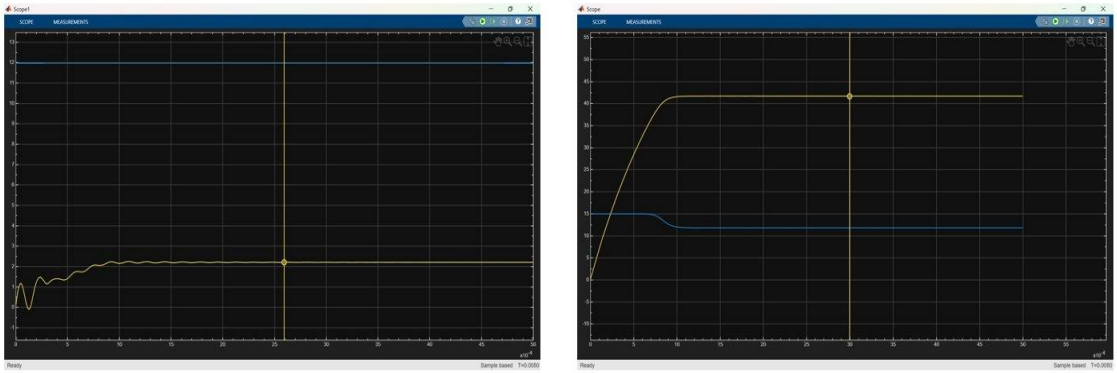


Figure 7. Simulink shuffler model with the inner phase-current PI loops in place. Each phase A and B has an independent PI controller driven by a fixed 3 A reference. The PWM signals are produced by per-phase gate-driver subsystems ("Phase A Gate Driver", "Phase B Gate Driver"). The coupled inductor is shown in the centre with the two phase-current sensors on its winding terminals.

2.10.5 Present Challenges and Future Work

The semester closes with one major technical blocker and two related items of unfinished design work. All three are concentrated in the three-loop cascaded architecture of 2.9: the inner loop is the active blocker, and the middle and outer loops have not yet been attempted in simulation.

Present challenge inner phase current loop. Closure of the inner phase current loop with a fixed reference, although qualitatively functional under uniform irradiance, becomes unstable under realistic partial shading. Figure 8 shows the symptom: a two-module test (500 W/m² on the top PV, 1000 W/m² on the bottom) with a fixed 3 A current reference. The bottom (illuminated) module settles cleanly to a steady V_pv ≈ 12 V and I_pv ≈ 2 A. The top (shaded) module, however, is driven by the controller to the open-circuit region its voltage saturates at approximately V_oc ≈ 42 V while its current collapses. Once the top module reaches open circuit it can no longer supply the 3 A demanded by the inner loop, and the controller cannot recover without an outer loop providing an operating-point-aware reference. The root cause is structural rather than tuning: a static current reference is not a meaningful target under mismatched irradiance because the controllable operating range of one module is a strong function of the other module's irradiance. The fix is the outer voltage loop, but the failure mode itself needs to be characterised and bounded before that work proceeds.



a) Bottom PV Module (1000 Irrad). Yellow is Vpv. Blue is Ipv.

b) Top PV Module (500 Irrad). Yellow is Vpv. Blue is Ipv.

* Note: Testing a 2 PV Modules with fixed current reference of 3A.

Figure 8. Inner phase current loop with fixed 3 A reference, two-module test, irradiances 1000 W/m² (bottom) and 500 W/m² (top). (a) Bottom module: V_pv (yellow) settles to ≈ 12 V, I_pv (blue) to ≈ 2 A stable. (b) Top module: V_pv (yellow) climbs into open-circuit saturation at ≈ 42 V while I_pv (blue) drops, demonstrating that a fixed current reference is not a valid operating target under partial shading.

Future work Loop 2 (voltage). The voltage loop has not yet been implemented. Its role is to command the inner-loop current reference dynamically so the per-module operating point tracks a desired V_C1 setpoint regardless of irradiance. With the voltage loop closed, the static-reference failure mode of Figure 8 should disappear by construction. The analytical gain prescription $K_{p,v} = \omega_{cv} \cdot C_1 / (2D)$ from 2.9 provides the seed values; tuning against the Simulink switching model is expected to refine them by typically 30–50%.

Future work Loop 3 (balance) and the missing application context. The balance loop has likewise not been implemented; it operates on the differential phase current and is structurally the simplest of the three. More important than either loop in isolation, the present model does not yet include the two elements that define the target use case: the MPPT outer loop (which sets V_C1 dynamically based on the partial-shading P-V landscape) and the battery load (which gives the string a well-defined I-V relationship rather than a stiff voltage-source approximation). Until these are added, the closed-loop behaviour observed in simulation is informative about the converter dynamics but is not yet reflective of the intended application. Both items are scheduled in the revised Semester 2 plan (Appendix A).

3. Risk Management

Risks have been actively monitored throughout the semester. The risk register is now dominated by a single high-impact item the project timeline with two related technical risks (electrical hazard, control–hardware integration) at lower current exposure. The proposal-era risk register remains the baseline; the discussion below identifies what has materialised, what has been mitigated, and what is now most likely to bite.

3.1 Critical Risk Project Timeline and PCB Order Delays

The most significant project change this semester is the postponement of the first PCB iteration (Shuffler V1 hardware) from Semester 1 into early Semester 2. The deferral consumed approximately three weeks of the one-month contingency buffer built into the original Gantt schedule. The remaining buffer is approximately one week small. Any further delay to the PCB order, component lead time, or first-fit on the hardware would exhaust the buffer and start displacing the late-semester milestones (M7 onwards), which were originally protected by the buffer. This is now the critical project risk and is being managed by tight weekly milestone review with the supervisor.

The decision to defer was made transparently and was the right call against the alternative committing to a hardware iteration with an incomplete control model would either have produced a non-functional first prototype or required a full redesign once the control work converged. But the cost of being right is that the schedule is no longer protected by buffer in the way it was at the start of the semester, and there is no second deferral available. Maintaining momentum on the Semester 2 deliverables (loops 2 and 3, QSPICE PI, MPPT integration, battery-load model) is now the dominant scheduling pressure.

3.2 Application-Realism Risk Model does not yet match target use case

The present Simulink model represents the shuffler operating into a controlled voltage / current-sweep load. The target end-use is a residential PV-battery system, in which the load is the battery's I-V characteristic (or, more precisely, the input of a battery-side DC-DC converter) and the operating point is set by an MPPT outer loop. Neither the battery load nor the MPPT loop is yet in the model. The closed-

loop behaviour observed in 2.10 is therefore informative about the converter dynamics but is not yet reflective of the intended application. This is a meaningful risk to flag now: control design choices that look correct against a stiff-source load may behave differently against a battery, and an MPPT outer loop introduces its own slow dynamics that can interact with the voltage loop. Adding both is on the Semester 2 critical path.

3.3 Other Risks

Electrical hazard (Risk #8, Critical)

All Semester 1 work has been performed in simulation. The QUT HSE workspace induction will be completed and PPE confirmed before the first hardware bench test. The PCB deferral has the side effect of giving additional time to prepare a formal hardware risk assessment before any live test.

Firmware failure (Risk #4, High)

Implementing the controller first in Simulink (where the plant model is fully observable and tunable) has surfaced several issues that would have been far harder to diagnose on hardware: dead-time interaction with body diodes, coupled-inductor model fidelity, and the inner-loop instability identified in 2.10.5. The simulation-first approach is working as designed and has been retained for Semester 2.

Control–hardware integration (Risk #13, High)

The QSPICE implementation of the PI controller has been started but is not yet operational. The current failure mode is a singular-matrix error during DC operating-point convergence. Resolving this before hardware deployment is the correct order of operations and is on the revised Semester 2 schedule.

Coupled inductor modelling fidelity

The current analysis treats the two inductors as independent (L_1 and L_2 separate). This is a defensible simplification for steady-state and average-current behaviour but understates the leakage-versus-magnetising-inductance dynamics of the real coupled inductor. The mitigation is to revisit the coupled-inductor model in MATLAB before the controller is transferred to QSPICE.

MPPT global tracking (Risk #14)

The partial-shading validation has confirmed the multiple-local-maxima problem visible in Figure 6: a naïve perturb-and-observe MPPT will trap at a local maximum under graduated shading. The MPPT design will need either a global search component or a topological argument that exploits the shuffler-equalised state. Carried forward into Semester 2.

4. Ethics

The project has been conducted in accordance with the Engineers Australia Code of Ethics and Guidelines on Professional Conduct. Four sections of the Code have been directly relevant to decision-making this semester. A fifth consideration the use of AI assistance in the analytical work is addressed in 4.5.

4.1 Demonstrate integrity Act on the basis of a well-informed conscience

The decision to postpone the PCB V1 order was taken under this principle. Producing hardware on the originally scheduled date with an incomplete control model would have created a high probability of

producing a non-functional or unsafe prototype. The decision to defer was made transparently with the supervisor and the hardware teammate and prioritises engineering integrity over schedule adherence.

4.2 Practise competently Maintain and develop knowledge and skills

The state-space methodology applied in 2 sits outside the undergraduate curriculum at this depth. The progression from topological enumeration through averaging, equilibrium, linearisation, stability/controllability/observability analysis, discretisation, and cascaded PI design has been undertaken with deliberate attention to first-principles correctness each result has been validated either against the underlying physics (energy conservation, sign of inductor voltage), against the literature (Batzelis et al. [10] for state-space methodology, Olalla et al. [5][6] for S-PPC behavioural expectations), or against the Simulink model.

4.3 Exercise leadership Engage with the relevant community and stakeholders

The two-person team structure (control systems and power electronics) requires active coordination on the interface between the two subsystems. Weekly supervisor meetings have been used to keep both halves of the project aligned, and the PCB postponement decision was made jointly to ensure that the hardware work is not blocked by a downstream control-design change.

4.4 Promote sustainability Engineer for a sustainable future

Detailed discussion in 5. The project's central technical objective recovering energy lost to partial shading mismatch is itself a sustainability outcome.

4.5 AI Assistance and the Explainability of Mathematical Work

AI tools (Anthropic's Claude, accessed via claude.ai, March–May 2026) were used to assist with the analytical derivations in 2 and Appendix B. This raises a question of professional responsibility that the Code of Ethics addresses indirectly under "practise competently": when a tool participates in the production of an engineering result, the engineer is responsible for the result's correctness regardless. The explainability question is therefore: can the student demonstrate, step by step, that each result is correct, and could a marker or supervisor verify each step independently?

Mathematics has a property that makes this question unusually tractable. Unlike, say, an opaque deep-learning model whose intermediate decisions are not human-readable, every step of a state-space derivation can be checked by substitution, by dimensional analysis, by limit cases ($D \to 0$ and $D \to 1$), and by numerical agreement with simulation. The work in Appendix B was developed in iterative dialogue with the AI tool but every result has been verified by at least one of these mechanisms before being included in the report. The matrices in Figure 2 were re-derived by hand. The equilibrium expression (Appendix B equation B.8) was numerically tested against the Simulink string model in 2.10.3. The eigenvalue result $\sigma(\tilde{A}) = \{0, 0, \pm j\omega_o\}$ was sanity-checked against the well-known marginal-stability behaviour of lossless converters from the literature. The observability rank deficiency was verified symbolically in MATLAB. None of this verification would be possible or would be much harder if the same work had been done in a domain where the underlying object resists hand-checking.

The practical implication for the project is that AI assistance has been used as a structural aid (proposing decompositions, surfacing equations to be checked, drafting the textbook-style presentation in Appendix B) rather than as a substitute for the student's own intellectual engagement with the derivation. The acknowledgement and a representative set of transcript excerpts are included in Appendix B B.13 for transparency.

5. Sustainability

Sustainability considerations have been applied using the Engineers Australia Implementing Sustainability: Principles and Practice framework. The project's primary contribution to sustainability is the recovery of PV string energy that would otherwise be lost to partial-shading mismatch a direct contribution to UN SDG 7 (Affordable and Clean Energy) by improving the efficiency of an already-installed renewable resource.

5.1 Whole-of-life thinking

The S-PPC topology was selected over the more common full-power DC-DC converter precisely because its component ratings scale with the differential (mismatch) power rather than the full string power. This has two life-cycle consequences: lower-rated, less material-intensive components are required for the same string size, and the converter dissipates less heat during normal operation, extending the effective component life. The decision to use the simplest practical S-PPC variant the interleaved half-bridge buck-boost rather than a more complex multi-level or flying-capacitor design (Schweizer & Soeiro [9], Jeong et al. [15]) is also driven by life-cycle considerations: fewer parts, lower failure surface, easier field repair.

5.2 Systems thinking

The control system is being designed to be topology-agnostic within the S-PPC family. The averaged state-space framework generalises across S-PPC variants by changing the per-state A matrices; the cascaded loop architecture and the modal decomposition carry across unchanged. This means the firmware developed in this project should remain useful as a building block for future S-PPC variants without a full redesign a sustainability dimension of the development effort itself, not just the end product.

5.3 Material and energy efficiency

Two specific decisions this semester have material or energy implications. First, the simulation-first development methodology has avoided the production of physical prototypes that would have been discarded once the control model was refined the PCB V1 postponement is, in this sense, a reduction in materials consumption. Second, the analytical bandwidth allocation (current loop at 10 kHz against a 200 kHz switching frequency) keeps the controller compute load modest, supporting the future deployment of the firmware on low-power microcontrollers rather than requiring a high-performance DSP.

6. Information Sources, Standards, and Resources

6.1 Academic literature

The analytical foundation has drawn principally on the state-space modelling work of Batzelis et al. [10], the S-PPC topology literature (Walker et al. [1], Kasper et al. [3][4], Olalla et al. [5][6], Jeong et al. [12][15], Dos Santos et al. [17]), and classical control texts underpinning cascaded PI design. The literature review carried out at project start remains current; no new references have been added this semester that materially change the design direction.

6.2 Standards and codes

- **MISRA-C 2012** the safety-critical coding standard targeted for the firmware. No firmware has been written yet, but the MATLAB controller code has been kept in a structure compatible with

MISRA-C constraints (no recursion, bounded loops, deterministic execution paths) to ease translation in Semester 2.

- **QUT HSE Electrical Workspace Protocols** not yet engaged this semester, but reviewed in preparation for the Semester 2 hardware tests.
- **IEEE conventions for control engineering documentation** adopted for figure labelling, state-vector notation, and matrix presentation.

6.3 Tools and software

The semester's work has used MATLAB / Simulink (R2024b) as the primary control design and simulation environment; QSPICE (2025 release) for switching-level circuit simulation; VS Code with PlatformIO in preparation for Semester 2 firmware work; GitHub (private repository) for version control and supervisor review; and OneDrive for file sharing with the hardware teammate.

7. Stakeholders

7.1 Supervisor Assoc Prof Geoff Walker

The supervisor has been engaged weekly throughout the semester. Some advice provided was the methodology choice to derive the averaged state-space model directly from the topology was endorsed at an early meeting and shaped the entire semester's analytical work.

7.2 Hardware teammate Son Nguyen

Son Nguyen leads the power electronics design within the team. The handover interface between the two halves of the project is the QSPICE schematic produced by Son, against which the control system PI will be implemented. This semester's interaction has centered on understanding the gate driver architecture (single-Vref-input design with hardware-enforced 180° interleaving via complementary carrier comparison) a hardware decision that constrains the control system to a single duty input per shuffler, which is consistent with the equilibrium constraint D_A = D_B derived in 2.5. No conflict has arisen between the two design directions; the interface assumptions are consistent.

8. Conclusion

Semester 1 has established the analytical foundation for the project and an initial Simulink implementation. The state-space model has been derived, linearised, analysed, and discretised. A cascaded PI controller has been designed analytically. The PV string and shuffler have been built in Simulink and validated against the analytical partial-shading predictions. Closure of the inner phase current loop has been attempted and has exposed a structural failure mode that defines the immediate next piece of work.

The framing for Semester 2 is therefore one of foundations laid and direction set, not delivery complete. The voltage loop, balance loop, MPPT integration, battery load model, and QSPICE bring-up are all unfinished. The PCB iteration has been deferred and the schedule no longer has the buffer it started with. The Semester 2 priority is to maintain momentum on each of these items so that the final-semester deliverables (validated closed-loop hardware demonstration, MPPT under partial shading, and the final report) remain achievable on the revised plan.

Appendix A Updated Timeline and Deliverables

The original Gantt chart prepared at project start remains the structural reference. Table A1 lists each deliverable from the original timeline and records its current status, with reasons for changes where applicable.

#	Phase	Deliverable	Original Wk	Status
1	Setup	Project Setup	1–2	Complete
2	Setup	Supervisor meeting & comms strategy	2	Complete (M1)
3	Planning	Project planning & literature review	3–7	Complete
4	Design	High-level system architecture	8	Complete (M2)
5	Design	Shuffler V1: circuit design & control model	9–12	Control model complete; circuit design deferred
6	Order	Shuffler V1 components ordered	10	Deferred to Sem 2 Wk 15
7	Design	Documented simulation models	12	Complete (M3, +1 wk)
8	Testing	Unit test plans & acceptance criteria	13–14	In progress
9	Testing	Shuffler V1 testing & experiment report	14	Deferred to Sem 2 Wk 16 (M4)
10	Order	Shuffler V2 & MPPT V1 ordered	15	Replaced see note below

Table A1. Status of original timeline deliverables at end of Semester 1.

A.1 Changes and Reasons

Change 1 Shuffler V1 hardware deferred

Item 6 (Shuffler V1 components ordered, original Wk 10) has been deferred to early Semester 2. The reason is the depth of the analytical modelling work required to support the control design: the state-space derivation, equilibrium analysis, and stability/controllability work consumed Semester 1 weeks 9–12, and the Simulink validation work consumed weeks 13 onwards. Committing to a PCB design at Wk 10 with an incomplete control model would have produced either a non-functional first iteration or required a full redesign. The deferral consumes approximately three of the four weeks of contingency buffer built into the original schedule.

Change 2 Item 10 reframing

The original Item 10 ("Shuffler V2 & MPPT V1 Ordered") was specified before the control model was developed. The semester's analytical work has established that the V1 control model is sufficient for both the original V1 and V2 use cases the same averaged model, the same cascaded architecture, only the operating-point parameters change. The hardware revision plan therefore becomes V1 (single shuffler), then V2 (pair), without a separate intermediate control redesign. This reduces the number of PCB iterations in Semester 2 from three to two and partially offsets the cost of Change 1.

Change 3 Additions

Two activities not in the original Gantt have been added:

- PV module characterisation in Simulink against the AIKO Neostar 3P60 datasheet (Wk 11–12).
- QSPICE behavioural-source PI implementation (Wk 14–15, in progress).

A.2 Revised Milestone Schedule

Milestones M1, M2, and M9–M10 are unchanged. M3 slips one week. M4 and M5 each shift later as a consequence of the PCB postponement. M6 and M7 shift correspondingly. M8 remains within the contingency buffer.

Milestone	Original Wk	Revised Wk	Notes
M3 Simulation models documented	12	13	+1 wk
M4 Shuffler V1 testing	14	17	+3 wk (PCB postponement)
M5 Shuffler V2 & MPPT V1 firmware	17	21	+4 wk
M6 Subsystem verification complete	19	23	+4 wk
M7 Shuffler V3 & MPPT V2 firmware	23	25	+2 wk (within contingency)
M8 System validated	27	28	+1 wk (within contingency)

References

Only entries cited by number in this report are reproduced here.

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Appendix B Fully Worked Analytical Derivation

This appendix presents the complete analytical derivation of the averaged state-space model of the interleaved half-bridge shuffler, its equilibrium and linearisation, and the closed-form analysis of stability, controllability, observability, and discretisation. Each step is presented in the form of a statement followed by the corresponding equation. Key equations are numbered (B.1)–(B.11) and are referenced from the body of the report.

Acknowledgement of AI-Assisted Development. The analytical derivations in this appendix were developed in iterative dialogue with Anthropic's Claude (Claude Opus model, accessed via claude.ai, March–May 2026). The role of the AI assistant was limited to: (i) structural assistance with the per-state KCL derivations of B.3; (ii) algebraic manipulation and intermediate-result checking through B.4–B.10; (iii) drafting of the textbook-style presentation; and (iv) typesetting. The student defined the problem statement, the converter topology, the state-variable choices, and the operating-point assumptions; validated each result against the underlying physics (energy conservation, volt-second balance, KCL/KVL sign consistency); cross-checked the equilibrium and stability results against the Simulink switching-model simulations reported in 2.10; and holds full intellectual responsibility for the correctness and interpretation of all results presented herein. Representative transcript excerpts documenting the collaborative development are provided in B.13.

B.1 Setup: State Variables and Sign Conventions

Define the state vector:

$$x = [I_L1,\ I_L2,\ V_C1,\ V_C2]^T \quad (B.1)$$

Control input vector (per-phase duty cycles):

$$u_d = [d_A,\ d_B]^T \quad (B.2)$$

Exogenous input vector (PV currents and load current):

$$u_e = [I_{pv1},\ I_{pv2},\ I_load]^T \quad (B.3)$$

Sign conventions are fixed as follows: I_Lk > 0 means current flows V_mid → L_k → switch node SW_k (k ∈ {1, 2}); V_C1 = V_top − V_mid (positive when top rail above midpoint); V_C2 = V_mid − GND = V_mid (positive when midpoint above ground); I_pv1 > 0 means internal current through X1 in direction V_mid → V_top; I_pv2 > 0 means internal current through X2 in direction GND → V_mid; I_load > 0 means current drawn from V_top and returned to GND.

B.2 Enumeration of Valid Switching States

Total switch combinations:

$$N_{total} = 2^4 = 16$$

Forbidden combinations correspond to shoot-through within either leg, i.e. any combination containing S1+S2 or S3+S4. Counting: 4 (S1·S2 alone) + 4 (S3·S4 alone) − 1 (both, double-counted) = 7 forbidden combinations.

Number of valid topological states:

$$N_valid = 16 - 7 = 9$$

The four dual-switch states (states 6–9) are the control-relevant operating modes. The single-switch and all-off states arise only transiently during dead time and are dynamically equivalent (via body-diode freewheel) to one of the four dual-switch states.

B.3 State-Space Derivation for Each Dual-Switch State

Each state has the form:

$$\dot{x} = A_i\, x + B\, u_e \quad i \in \{6, 7, 8, 9\} \quad (B.4)$$

The input matrix B is identical across all four states since the exogenous inputs enter the capacitor nodes independently of switch position. Only A_i varies.

B.3.1 State 9 (S1 + S3 ON) both high-side

Switch node clamps: V_SW_A = V_top, V_SW_B = V_top. Inductor voltages by KVL: V_L1 = V_mid − V_top = −V_C1, V_L2 = −V_C1. KCL at V_top gives I_C1 = I_pv1 + I_L1 + I_L2 (no load in the analytical form; load added in B.5). KCL at V_mid simplifies to I_C2 = I_pv2. Collected:

$$L_1\, dI_L1/dt = -V_C1$$

$$L_2\, dI_L2/dt = -V_C1$$

$$C_1\, dV_C1/dt = I_L1 + I_L2 + I_{pv1}$$

$$C_2\, dV_C2/dt = I_{pv2}$$

B.3.2 State 6 (S2 + S4 ON) both low-side

Switch node clamps: V_SW_A = 0, V_SW_B = 0. Inductor voltages: V_L1 = +V_C2, V_L2 = +V_C2. Applying the same KCL procedure:

$$L_1\, dI_L1/dt = +V_C2$$

$$L_2\, dI_L2/dt = +V_C2$$

$$C_1\, dV_C1/dt = I_{pv1}$$

$$C_2\, dV_C2/dt = -I_L1 - I_L2 + I_{pv2}$$

B.3.3 States 7 and 8 (diagonal)

States 7 (S2 + S3) and 8 (S1 + S4) are the diagonal configurations. By the same procedure: in State 8, Phase A is high-clamped and Phase B is low-clamped, giving V_L1 = −V_C1, V_L2 = +V_C2, with capacitor equations $C_1\,\dot{V}_C1 = I_L1 + I_pv1$ and $C_2\,\dot{V}_C2 = -I_L2 + I_pv2$. State 7 mirrors State 8 by swapping Phase A and B roles.

B.4 Time-Averaged Model

Assume continuous conduction mode and independent phase switching. The four dual-switch states have dwell fractions d_A d_B, d_A(1−d_B), (1−d_A) d_B, and (1−d_A)(1−d_B) over one switching period. Sum of dwell fractions equals 1 by inspection. The averaged A matrix is the dwell-weighted sum:

$$\bar{A} = d_Ad_B\, A_9 + d_A(1-d_B)\, A_8 + (1-d_A)\, d_B\, A_7 + (1-d_A)(1-d_B)\, A_6 \quad (B.5)$$

Carrying out the weighted sum row by row gives the averaged nonlinear model presented in 2.4 of the main body. The two inductor equations are decoupled each phase's average dynamics depends only on its own duty. The capacitor equations couple both phases through the sum of duty-weighted currents.

B.5 Equilibrium Operating Point

At equilibrium, all time derivatives vanish. From the two inductor equations: D_A V̄_C1 = (1−D_A) V̄_C2 and D_B V̄_C1 = (1−D_B) V̄_C2. Equating gives D_A / (1−D_A) = D_B / (1−D_B). Since D/(1−D) is strictly monotonic on (0, 1):

$$D_A = D_B \equiv D \quad (B.6)$$

Both phases must run at the same steady-state duty. The differential duty (d_A − d_B) is a transient degree of freedom only. Substituting D_A = D_B = D into either volt-second equation gives the steady-state voltage ratio:

$$\bar{V}_C2/\bar{V}_C1 = D/1 - D \quad (B.7)$$

Adding the load current to the V_top KCL (as physically required, since the no-load case yields the degenerate equilibrium with I_pv → 0) and solving the resulting charge-balance system for D:

$$D = I_load - I_{pv1}/I_load - I_{pv1} + I_{pv2} \quad (B.8)$$

Per-phase steady-state current (symmetric split at the nominal operating point):

$$\bar{I}_L1 = \bar{I}_L2 \equiv \bar{I} = I_{pv2}/2(1 - D) \quad (B.9)$$

B.6 Linearisation About the Operating Point

Define perturbations x = X̄ + x̃, d = D + d̃, u_e = Ū_e + ũ_e. Substituting into each averaged equation and retaining first-order terms (using bilinear product expansion XY ≈ X̄Ȳ + X̄·ỹ + Ȳ·x̃) yields the linearised model x̃̇ = Ã·x̃ + B̃_d·d̃ + B̃_e·ũ_e. The matrices Ã, B̃_d, B̃_e are presented in Figure 2 of the main body; Ã depends on the operating-point duty D and the component values $L_{1,2}$, $C_{1,2}$. B̃_d depends additionally on V_Σ = V̄_C1 + V̄_C2 and Ī the actuator effectiveness changes with bus voltage and current.

B.7 Eigenvalues and Stability

Assume $L_1 = L_2 = L$ for closed-form analysis. Rows 1 and 2 of Ã are identical (both phases see the same V_C1, V_C2) and columns 1 and 2 of Ã are identical (both phases contribute identically to the capacitors). Each rank deficiency yields one zero eigenvalue. Applying the modal change of variables I_Σ = I_L1 + I_L2 and I_Δ = I_L1 − I_L2: the differential dynamics decouple into a pure integrator with eigenvalue 0; the common-mode subsystem in state (I_Σ, V_C1, V_C2) has characteristic polynomial $\lambda^3 + \lambda[2D^2/(LC_1) + 2(1-D)^2/(LC_2)] = 0$, factoring as:

$$\lambda\,[\lambda^2 + {\omega_0}^2] = 0, \quad {\omega_0}^2 = 2D^2/LC_1 + 2(1 - D)^2/LC_2$$

Combined with the differential-mode zero, the full eigenvalue set is:

$$\sigma(\tilde{A}) = \{\, 0,\ 0,\ +j\omega_0,\ -j\omega_0 \,\} \quad (B.10)$$

Two eigenvalues at the origin, one undamped imaginary pair. The system is marginally stable. Numerical substitution (L = 8.2 µH, $C_1 = C_2 = 100$ µF, D = 0.5) gives $\omega_0 \approx 34\,900$ rad/s, $f_0 \approx 5.55$ kHz.

B.8 Controllability

Test the pair (Ã, B̃_d) via the modal decomposition. The 1-D differential subsystem has L · dĨ_Δ/dt = −V_Σ · ũ_δ reachable from a single input ũ_δ provided V_Σ ≠ 0, so rank [B_Δ, A_Δ B_Δ] = 1 and the differential mode is controllable. The 3-D common-mode subsystem under input ũ_c = (d̃_A + d̃_B)/2 has input matrix

$B_c = [-2V_\Sigma/L, 2\bar{I}/C_1, 2\bar{I}/C_2]^T$; all three entries are non-zero when $V_\Sigma \neq 0$ and $\bar{I} \neq 0$, and the controllability matrix $[B_c, A_{red} B_c, A_{red}^2 B_c]$ is generically rank-3. The plant is therefore fully controllable at all non-degenerate operating points. Controllability is lost only at $\bar{I} \to 0$ (DCM idle), $D \to 0$ or $D \to 1$ (saturated duty), or $V_\Sigma \to 0$ (physically unreachable).

B.9 Observability

Observability depends on the output matrix C. With C = [V_C1, V_C2] only (rows [0,0,1,0] and [0,0,0,1]), propagating through $\tilde{A}$ every row of the observability matrix depends on (I_L1 + I_L2) only never on (I_L1 − I_L2). Therefore rank(O) = 3 < 4 and the system is NOT observable; the differential current is hidden. Adding I_L1 to the measurement set recovers I_L2 by subtraction (V_C1 dynamics yields the sum (I_L1 + I_L2)/C_1), giving rank(O) = 4 and full observability. With C = I_4 (full state measurement) observability is trivial. Per-phase current sensing is mandatory for interleaved operation. Voltage-only observability is structurally insufficient.

B.10 Discretisation

Sampling period: $T_s = 1/f_{sw} = 1/200$ kHz = 5 µs. Zero-order hold discretisation (exact for piecewise-constant duty): $A_d = \exp(\tilde{A} \cdot T_s)$, $B_d = \int_0^{T_s} \exp(\tilde{A}\tau)\, d\tau \cdot \tilde{B}$. Because $\tilde{A}$ is singular, the integral form is required and evaluated via the augmented exponential trick. Discrete eigenvalues map as $\lambda_d = \exp(\lambda_c \cdot T_s)$: $\lambda_c = 0 \to \lambda_d = 1$ (×2); $\lambda_c = \pm j\omega_o \to \lambda_d = \cos(\omega_o T_s) \pm j \sin(\omega_o T_s)$. Numerical ($\omega_o T_s \approx 0.175$ rad): λ_d (oscillatory) ≈ 0.985 ± j 0.174, $|\lambda_d| = 1$. All four discrete eigenvalues sit on the unit circle, mirroring the continuous-time marginal stability.

B.11 Analytical PI Gain Selection

Inner phase current loop. Plant (V_Σ held constant by the slower voltage loop) is a pure integrator $G_{id}(s) = -V_\Sigma/(sL)$. PI controller $C_i(s) = K_{p,i}(\tau_i s + 1)/(\tau_i s)$. Setting $|L_i(j\omega_{ci})| = 1$ at the target crossover ω_{ci}:

$$K_{p,i} = \omega_{ci} L / V_\Sigma, \quad \tau_i = 10/\omega_{ci}$$

Outer voltage loop (assuming the closed inner loop is ideal). Plant $G_{vi}(s) \approx 2D/(sC_1)$. By the same procedure:

$$K_{p,v} = \omega_{cv} C_1 / 2D$$

Balance loop. Same plant structure as the inner current loop, operating on $\tilde{I}_\Delta$. $K_{p,\delta} = \omega_{c,\delta} L / V_\Sigma$ with $\tau_\delta = 10/\omega_{c,\delta}$. All three loops achieve phase margin ≈ 84° from the placement of the PI zero one decade below crossover.

B.12 Summary of Key Analytical Results

Result	Expression	Eq.
Steady-state duty relation	$D_A = D_B \equiv D$	(B.6)
Voltage ratio at equilibrium	$\bar{V}_{C2} / \bar{V}_{C1} = D / (1 - D)$	(B.7)
Steady-state duty (with load)	$D = (I_{load} - I_{pv1}) / (I_{load} - I_{pv1} + I_{pv2})$	(B.8)
Per-phase steady current	$\bar{I} = I_{pv2} / [2(1 - D)]$	(B.9)
Open-loop eigenvalues	$\sigma(\tilde{A}) = \{ 0, 0, \pm j\omega_o \}$	(B.10)

Result	Expression	Eq.
Natural resonance	$\omega_o^2 = 2D^2/(LC_1) + 2(1-D)^2/(LC_2)$	
Controllability	Full at all non-degenerate operating points	
Observability (V_C only)	rank(O) = 3 differential current hidden	
Observability (V_C + I_L1)	rank(O) = 4 full state observable	

B.13 Documentation of AI-Assisted Development

This section provides representative excerpts from the dialogue with Claude that underpinned the analytical work. The excerpts have been selected to demonstrate the dialectic nature of the collaboration: the student set the problem and the constraints, the AI assistant proposed structural formulations and algebraic manipulations, and the student validated each result before it was carried forward.

Excerpt 1 Setting up the per-state averaging framework

Context: Early in the modelling work, the choice between (a) deriving an averaged model directly from a generic buck-boost prototype and modifying it, or (b) enumerating every valid switching state of the interleaved half-bridge and averaging from first principles, was discussed. The decision to take the first-principles route shaped all subsequent analytical work and is reflected in B.3 of this appendix.

> **Student:** [summarising the topology setup] … "Now I believe I have identified 9 topological layouts based on the operation of the switches. 1. all switches are off; 2. switch 1 is on; 3. switch 2 is on; 4. switch 3 is on; 5. switch 4 is on; 6. switches 1,3 are on (top); 7. switches 2,4 are on (bottom); 8. switches 1,4 are on (diag); 9. switches 2,3 are on (diag). Can you validate this approach and help me step through determing the ss model of each topological state so I can create an averaged ss model?"
>
> **Claude:** "Your enumeration looks correct and complete. You have 4 switches, giving $2^4 = 16$ total combinations. Eliminating the shoot-through cases (S1+S2 on, or S3+S4 on, which would short C_1 or C_2 through the leg) … that removes 7 of the 16 combinations, leaving 9 valid topological states matching your list exactly. Your list is well-organized too: zero-switch, single-switch (×4), then dual-switch (top-pair, bottom-pair, two diagonals). [Claude then proceeded to derive each state matrix; the four dual-switch states became the basis of the averaged model in B.3–B.4.]"

Student validation: The proposed averaging structure was checked by hand on State 9 and State 6 independently, confirming the symmetry of the four dual-switch states and the identical B matrix asserted in B.3.

Excerpt 2 Discovering the load dependence of the equilibrium duty

Context: Initial attempts to solve for the steady-state duty produced a degenerate result. The discussion clarified that a finite load current is required for a non-trivial equilibrium; this constraint is now made explicit in (B.8) of the appendix.

> **Student:** "lets smash out the equilibrium point analysis and linearisation"
>
> **Claude:** [after setting the time derivatives to zero and deriving D_A = D_B ≡ D and the voltage ratio $\bar{V}_{C2}/\bar{V}_{C1} = D/(1-D)$, the response continued]… "Step 5: We've hit a problem. Equation (*) cannot be satisfied for any $D \in (0,1)$ when both I_pv1 and I_pv2 are positive and PV photocurrents are always positive in normal generation. This isn't an algebra error it's a physical limitation of the model as written. KCL at V_top: PV X1 injects I_pv1 into the node. With C_1 at steady state (no displacement current), where does that current go? The only paths are the HS switches into the inductors … both PV currents are trying to leave their respective nodes via the inductors, but the inductors can only carry current in one direction at a time on average. So the system cannot find a steady state with both PVs generating positive power there's no sink to receive the energy. In a real installation, the PV string has a load … Step 6: Add a load model. The

simplest fix: add a load drawing constant current I_load from V_top, returning to GND. [Claude then solved for the modified equilibrium, yielding the expression for D given in (B.8).]"

Student validation: The closed-form duty expression was numerically tested against the Simulink string model under three partial-shading scenarios (2.10.3 of the main report); the predicted duty agreed with the simulation steady state to within the expected averaging error.

Excerpt 3 Observability finding (voltage-only measurement insufficient)

Context: A non-obvious result that materially affected the sensor specification was the rank deficiency of the observability matrix when only the two capacitor voltages are measured. This finding made per-phase current sensing a hard design requirement and is documented in B.9.

Student: "just before we do equilibrium point analysis and linearisation should we assess the stability, controllability and observability of the system in continuous time"

Claude: "Good call this should come before linearization. These properties tell us what's possible before we design anything. [The response derived the eigenvalues $\sigma(\tilde{A}) = \{0, 0, \pm j\omega_o\}$, established generic controllability, then turned to observability]… With only V_C1 and V_C2 measurements (no current sensors): the differential mode (I_L1 − I_L2) is unobservable. The two capacitor voltages only see the sum I_L1 + I_L2. Any common-mode current is reflected in V_C1 and V_C2 dynamics, but a pure differential current (I_L1 increases by Δ, I_L2 decreases by Δ) leaves both capacitor voltages unchanged. You can't detect current imbalance between phases without a current sensor. This is important for your design: at minimum you need one phase current sensor to detect and correct phase current imbalance, otherwise the differential mode drifts unboundedly under perturbations. This is a hard requirement for interleaved topologies."

Student validation: The rank-deficiency claim was independently verified by computing rank(O) symbolically in MATLAB for the linearised plant at the nominal operating point. The minimum-viable sensor set (V_C1, V_C2, I_L1) was confirmed sufficient and has been adopted as the hardware requirement for the Semester 2 PCB iteration.

Note on completeness

The three excerpts above are representative rather than exhaustive. The full set of transcripts is retained by the student and is available for inspection by the supervisor or markers on request. Where the AI assistant produced material that did not survive student validation (for example, an early controller gain expression that was later corrected through frequency-domain analysis), that material does not appear in this appendix and is documented separately in the project log.

End of Appendix B.